P–C–458a
APRIL 1, 1954
SUPERSEDING
Fed. Spec. P–C–458
September 13, 1946

FEDERAL SPECIFICATION

CLOTH, ABRASIVE, CROCUS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—Crocus cloth covered by this specification is a cloth uniformly coated on one surface with crocus. It is used to produce a high polish on metal.

1.2 Classification.

1.2.1 *Type.*—The crocus cloth covered by this specification shall be of one type.

1.2.2 *Form.*—Crocus cloth shall be furnished in sheet and roll form as specified.

1.2.3 *Dimensions.*—Dimensions of sheets and rolls, as indicated in 3.2, shall be as specified.

2. APPLICABLE SPECIFICATIONS AND STANDARD

2.1 The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

CCC–T–191—Textile Test Methods.

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Non-Government activities may obtain copies of Federal Specifications and Standards upon application accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets SW., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Copies of the Index of Federal Specifications and Standards may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices.

(Single copies of this specification and other product specifications required by non-Government activities for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards through departmental channels from the established sources of supply.

(Copies of Military Standard referenced above, required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Specifications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Abrasive.*—The abrasive shall consist essentially of ferric oxide (Fe_2O_3) and quartz. (See 3.4.)

3.1.2 *Backing.*—The backing material shall be a good quality jeans cloth as specified hereinafter.

3.1.3 *Adhesive.*—The adhesive shall be a good quality, animal-hide glue base, suitable for the purpose intended. (See 3.3.)

3.2 Dimensions of sheet and rolls.—Unless otherwise specified, crocus cloth shall be furnished in sheets not less than 9 x 11 inches, and in rolls not less than 50 yards long and in widths as specified.

3.3 Backing.—The backing shall be properly dyed and backfilled preparatory to coating. The grey cloth shall conform to the requirements shown in table I, when tested as described in 4.3.1. (See 3.4.)

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TABLE I.—*Backing requirements*

Yards per pound [1]	Thread count per inch	
	Warp	Filling
3.54 yards ±5 percent	96+5 −3	64+5 −3

[1] Nominal yards per pound of grey cloth is on a 30-inch-wide basis.

3.4 Fineness of abrasive.—The abrasive shall be of the following fineness:

	U. S. Standard Sieve No.
All through	140
Not more than 20 percent retained on	325

3.5 Breaking strength.—The average breaking strength of crocus cloth per inch width shall be not less than 100 pounds in the longitudinal direction, and 50 pounds in the transverse direction, when tested as described in 4.4.3.

3.6 Flexibility.—There shall be no visual evidence of cracking in the abrasive coating or flaking of the abrasive coating from the cloth backing, when tested as described in 4.4.4.

3.7 Identification marking.—Each sheet shall be marked on the back in a legible manner with the manufacturer's name or trade-mark of such character that the source of manufacture may be readily determined, and with the name of the product. At least every 3 feet of roll cloth shall be marked in a manner similar to the sheet.

3.8 Workmanship.—The material shall be finished in a workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—For purposes of sampling a lot shall consist of all crocus cloth presented at one time.

4.1.2 *Sampling for inspection of cartons.*—A random sample of crocus cloth shall be taken from each lot by the Government inspector in accordance with Military Standard MIL-STD-105 at Inspection Level I and Acceptable Quality Level equal 2.5 percent defective to verify compliance with the requirements of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.1.3 *Sampling for lot acceptance.*—From each of the sample cartons chosen in accordance with 4.1.2, two sheets of crocus cloth shall be chosen at random, no two sheets from the same folder, and shall be subjected to visual and dimensional inspection and tests with lot acceptance based on sampling requirements of table II. If the sample cartons contain rolls of cloth rather than sheets then 2 square yards of cloth shall be chosen, each yard from a separate roll except where the carton contains only one roll.

TABLE II.—*Sampling for tests*

Sample cartons	Sample sheets or square yards	Acceptance number, defective	Rejection number, defective
2	4	0	1
3	6	0	1
5	10	1	2
7	14	1	2
10	20	2	3
15	30	3	4
25	50	4	5

4.2 Inspection.

4.2.1 *Inspection of cartons.*—Each sample filled carton selected in accordance with paragraph 4.1.2., shall be examined by the Government inspector for defects of the carton and the closure and for unsatisfactory markings; each sample carton shall also be weighed to determine the amount of the contents. Any carton in the sample having one or more defects or under required weight shall be rejected and if the number of defective cartons in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard 105 the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided that the contractor has removed or repaired all nonconforming cartons.

4.3 Lot acceptance tests.—The sample specimens selected in accordance with 4.1.3 shall be visually and dimensionally inspected to ver-

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ify compliance with this specification and shall be subjected separately to the tests specified in paragraph 4.4.

4.4 Tests.

4.4.1 *Cloth-backing test.*—A 1-foot-square sample of the grey cloth backing shall be conditioned at 70° F.±2° F. and 65 percent ±2 percent relative humidity for not less than 12 hours and tested in accordance with method 5040 of Federal Specification CCC–T–191 for determination of weight, and method 5050 for determination of thread count.

4.4.2 *Fineness of abrasive.*

4.4.2.1 *Test specimen.*—Prepare one test specimen 1 inch x 9 inches (9 square inches) and one specimen 90 square inches in area from a sheet or roll of the crocus cloth under test.

4.4.2.2 *Weight of abrasive.*

4.4.2.2.1 *Centrifuge method.*—Immerse the 1-inch x 9-inch specimen in a 250-milliliter beaker containing 50 milliliters of distilled water. Cover with a watch glass and heat until the abrasive is free from the cloth. Remove the cloth and wash the adhering abrasive back into the beaker with a jet of hot water. Boil the content of the beaker until the volume is reduced to approximately 50 milliliters and transfer to a weighed 100-milliliter, round-bottomed, centrifuge tube with pour out. Place the tube in a centrifuge and centrifuge at approximately 1,500 revolutions per minute until the liquid is clear and the abrasive has settled. Pour off the clear liquid; add 40 milliliters of hot water to the tube and stir the settled abrasive. Centrifuge three times as prescribed above. After the third centrifuge, decant the liquid and dry the tube and abrasive in an oven at 100° C. Weigh the abrasive to the nearest milligram and calculate the weight of the abrasive.

4.4.2.2.2 *Filtration method (alternate).*—Immerse the 1-inch x 9-inch specimen in water in a 250-milliliter beaker, cover with a watch glass, and heat until the abrasive is free from the cloth. Remove the cloth and wash the adhering abrasive back into the beaker with a jet of hot water. Filter the contents of the beaker through a weighed No. 590 S & S White Ribbon (or a No. 40 Whatman) filter paper and wash thoroughly with warm water. Dry the filter paper and residue (or ignite in a platinum crucible). Weigh the abrasive to the nearest milligram and calculate the weight of the abrasive.

4.4.2.3 *Particle size of abrasive.*—Cut the 90-square-inch specimen in convenient pieces for immersion in water in an 850-milliliter beaker, cover with a watch glass, and heat until the abrasive is clear of the cloth. Remove the cloth and wash the adhering abrasive back into the beaker with a jet of hot water. Pour the contents of the beaker on a No. 140 sieve stacked on a No. 325 sieve and wash thoroughly through each sieve. Dry the sieves and abrasive and weigh the abrasive on each sieve to the nearest milligram. Calculate the amount of abrasive originally present on the 90-square-inch test specimen, assuming the weight of the abrasive per square inch to be the same as for the 9-square-inch specimen as determined by 4.4.2.2.1 and 4.4.2.2.2. The abrasive on each sieve may then be expressed in percent of the total abrasive.

4.4.3 *Breaking strength test.*—The breaking strength shall be determined by the cut-strip method in accordance with method 5102 of Federal Specification CCC–T–191, except that the jaws of the tension testing machine shall be set 5 inches apart and except that 6 specimens 1 inch wide shall be cut both longitudinally (direction of warp) and transversely (direction of fill) from the sample sheet material, and 6 specimens 1 inch wide shall be cut longitudinally (direction of warp) from the sample roll material. The test specimens shall be conditioned for 12 hours at 70° F.±2° F. and 50 percent±2 percent relative humidity. The average breaking strength shall be computed from at least 3 specimens. Samples tearing at the jaws of the tensile machine shall be discarded.

4.4.4 *Flexibility.*—At least five flexibility test pieces 1 inch in width shall be cut longitudinally from samples taken at random from the material under inspection. The specimens shall be flexed with the ends parallel over a $\frac{3}{16}$-inch smooth steel rod securely held in a vise or other suitable means, with the abrasive side away from the rod. They shall be given 20 to-and-fro motions of approximately 3 inches,

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under tension of about 30 pounds at a temperature of 70° to 77° F. Specimens shall be inspected for evidence of cracking in the abrasive coating or flaking of this coating from the cloth backing as a result of these tests.

4.4.5 Material represented by samples shall be accepted or rejected in its entirety based on the results of tests of the representative samples.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 *Sheet form.*—The crocus-cloth sheets may be packaged in either chipboard sleeves or folders. The number of sheets per sleeve or folder and the number of sleeves or folders per carton shall be as follows:

Sheets per sleeve or folder	Sleeves or folders per carton	Number of sheets per carton
50	10	500

5.1.1.1 *Fiberboard cartons.*—The fiberboard cartons used to package sheet form shall be in compliance with Federal Specification LLL–B–631 or LLL–B–636. Closure of the fiberboard boxes shall be made with adhesive, metal fastenings, tape, or a combination of these methods, in such a manner that the closure of the boxes shall not break or open when the box is tested in accordance with the applicable container specification.

5.1.2 *Roll form.*—Unless otherwise specified, roll crocus cloth shall be carefully wrapped in strong paper, and packed in standard commercial containers or wrapping, according to the packing schedule shown below, and shall be prepared for shipment to permit acceptance by carrier for transportation at lowest possible rate and afford maximum protection from normal hazards of transportation.

5.2 Marking.—Individual packages and shipping containers shall be marked with the name of the material, and the quantity contained therein, as defined by the contract or order number under which shipment is made, and the name of the manufacturer. In addition, shipping containers shall be marked with the contract or order number.

Packing schedule

Width of rolls	Rolls per package
Inches	
¾ and narrower	20
1–2½	10
3–3½	8
4–5	6
6–7	4
8–9	3
10–15	2
16–36	1

6. NOTES.

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and should specify width if rolls are desired (see 3.2.1.1, and 5.1.2).

6.2 Maximum work value.—Coated abrasives should be stored in a cool, even temperature where the relative humidity is somewhere between 35 and 65 percent; best work value will be obtained ordinarily if stored and worked in rooms having a relative humidity of 50 percent. It will give a work value of only about 50 percent if used in a 70 percent relative humidity. Coated abrasives should not be stored on a cement floor.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 Transportation description.—The proper transportation description, with applicable carload and truckload weights, is as follows:

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Abrasive Cloth.
Carload weight 36,000 pounds.
Truckload weight 22,500 pounds.

Patent notice.—When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specification of the issue in effect on date of invitations for bids, and special requirements, forms a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications.*

Military Specification:

MIL–P–3816—Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 Crocus cloth shall be packaged, packed and marked in accordance with Military Specification MIL–P–3816.

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Air Force.

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